

SECTION A — CONCEPT APPLICATION

S1. Compatible Motherboard, CPU, RAM and Storage

For a new office PC, I would first decide the CPU according to the work: Microsoft Office, web browsing, accounting software, video meetings and multitasking. Then I would choose a motherboard that supports the CPU socket, CPU generation and chipset. After that, I would select RAM that matches the motherboard, such as the correct DDR generation and supported speed. Finally, I would choose storage based on the available M.2/NVMe or SATA interface and required capacity.

- Check CPU socket and supported CPU generation.
- Check motherboard chipset and BIOS/UEFI support.
- Match RAM generation (for example DDR4 or DDR5), speed and capacity.
- Check whether storage uses M.2/NVMe or SATA and confirm the motherboard has the required connector.
- Before buying, verify the manufacturer's specifications and CPU support list.

S2. HDD vs SSD vs NVMe

An HDD uses spinning magnetic disks. It gives large capacity at a low cost, but it is slower. A SATA SSD uses flash memory and has no moving parts, so booting and application loading are much faster. An NVMe SSD also uses flash memory, but it communicates over PCIe, giving higher bandwidth and lower latency than a traditional SATA SSD.

- HDD: good for low-cost bulk storage, but slower.
- SATA SSD: faster and quieter than HDD.
- NVMe SSD: fastest of the three for normal modern PC storage.
- For office work, an NVMe SSD is a strong choice because it improves boot time, application loading and general responsiveness.

S3. GPU and Cooling

The GPU handles graphics such as images, 3D models and visual effects. For graphics-intensive design work, a suitable GPU can reduce the graphics workload placed on the CPU. The cooling system uses fans, heatsinks and thermal material to move heat away from the CPU and GPU.

- A GPU improves rendering and graphics performance.
- Cooling keeps temperatures within a safe range.
- If temperatures become too high, the system can reduce performance or shut down.
- Good airflow helps maintain stable performance during long work sessions.

S4. Why Hardware Compatibility Matters

Every component must physically fit and electrically communicate with the others. Compatibility includes socket type, RAM generation, storage interface, power connectors, case size and BIOS support.

- Wrong CPU socket → CPU cannot be installed or used.
- DDR4 RAM cannot be installed in a DDR5-only motherboard.
- A storage device may not work if its required interface is unsupported.

- An undersized PSU may not provide enough power.
- A component may not physically fit the case.
- An older BIOS may not support a newer CPU.

S5. Choosing and Connecting the PSU

Choose a PSU after estimating the power needed by the CPU, GPU, motherboard, storage and other devices. Leave some extra capacity for future upgrades and check PSU quality and efficiency. Before powering on, keep the PSU switched off and unplugged while checking every connection.

- Connect the 24-pin motherboard power cable.
- Connect the 4/8-pin CPU power cable.
- Connect the required PCIe power cable to a dedicated GPU.
- Connect SATA power to SATA drives if used.
- Keep loose cables away from fans and check for loose screws or possible shorts.
- Use the PSU's original cables; do not mix modular cables from different PSU models.

S6. POST Check and Troubleshooting

After assembly, power on the PC and check that the display appears and the system reaches BIOS/UEFI. Confirm that CPU, RAM, storage and GPU are detected. Watch for beep codes, diagnostic LEDs or error messages.

- If POST fails, switch the PC off first.
- Check the PSU and main power connections.
- Check 24-pin motherboard and CPU power cables.
- Reseat RAM and check for correct detection.
- Check CPU installation and compatibility.
- Check GPU and storage connections.
- Use BIOS/UEFI, beep codes or diagnostic LEDs to identify the problem, then test POST again.

Quick Memory Tip

CPU = works on instructions | RAM = temporary working space | Storage = long-term files | GPU = graphics | PSU = power | Cooling = removes heat | Motherboard = connects everything.

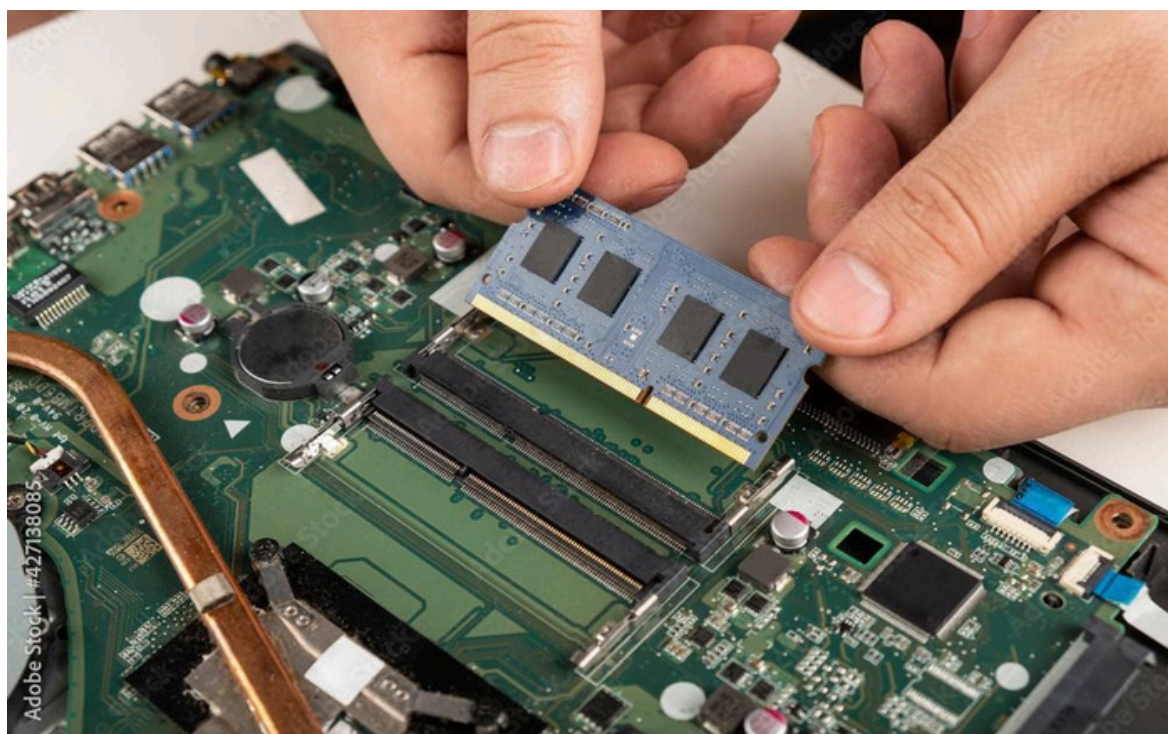
SECTION B — PRACTICAL TASKS

Task 1 — Component Identification & Functions

Component	Easy function	Key specification / example
Motherboard	Connects the CPU, RAM, storage, GPU and other hardware	Motherboard / chipset / sockets
CPU	Processes instructions and calculations.	Intel Core i5-1235U, 12th Gen (from assessment)
RAM	Temporarily holds data and programs being used.	16 GB (from assessment)
GPU	Processes and renders graphics.	Intel UHD Graphics (from assessment)
Storage	Permanently stores Windows, apps and files.	477 GB shown in Windows (from assessment)
Cooling	Removes heat and keeps components safe.	Cooling fan

Task 2 — Motherboard Ports & Socket Identification

Port / Part	Easy purpose
CPU Socket	Holds and connects the CPU to the motherboard.
24-pin ATX Power	Main power connection for the motherboard.
8-pin CPU Power	Provides power to the CPU.
SATA Ports	Connect SATA HDDs, SSDs and optical drives.
PCIe x16 Slot	Main slot used for a dedicated graphics card.
RAM / DIMM Slots	Hold desktop RAM modules.
USB Headers	Connect front-panel USB ports.
CPU Fan Header	Connects the CPU cooling fan.
M.2 Slot	Used for compatible M.2 NVMe SSDs.
PCIe x1 Slot	Used for smaller expansion cards such as Wi-Fi or sound cards.

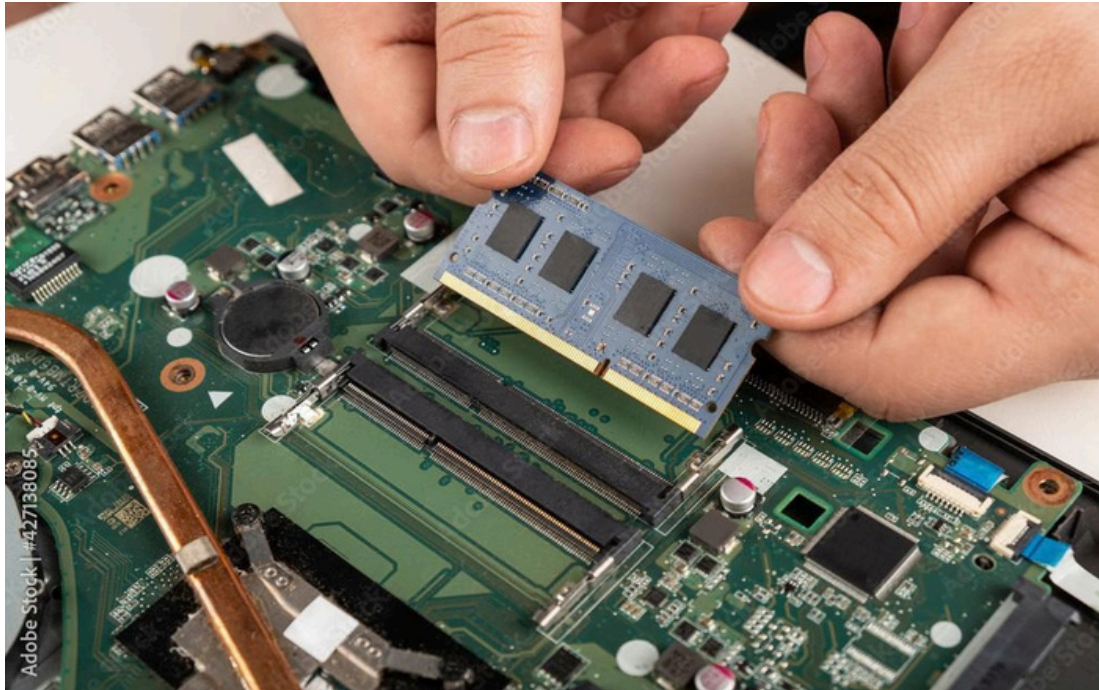


Real motherboard/port visual from the supplied assessment. Use it to identify CPU socket, RAM slots, SATA ports, PCIe slots and power connectors.

Task 3 — RAM & Storage Installation

RAM installation — easy steps

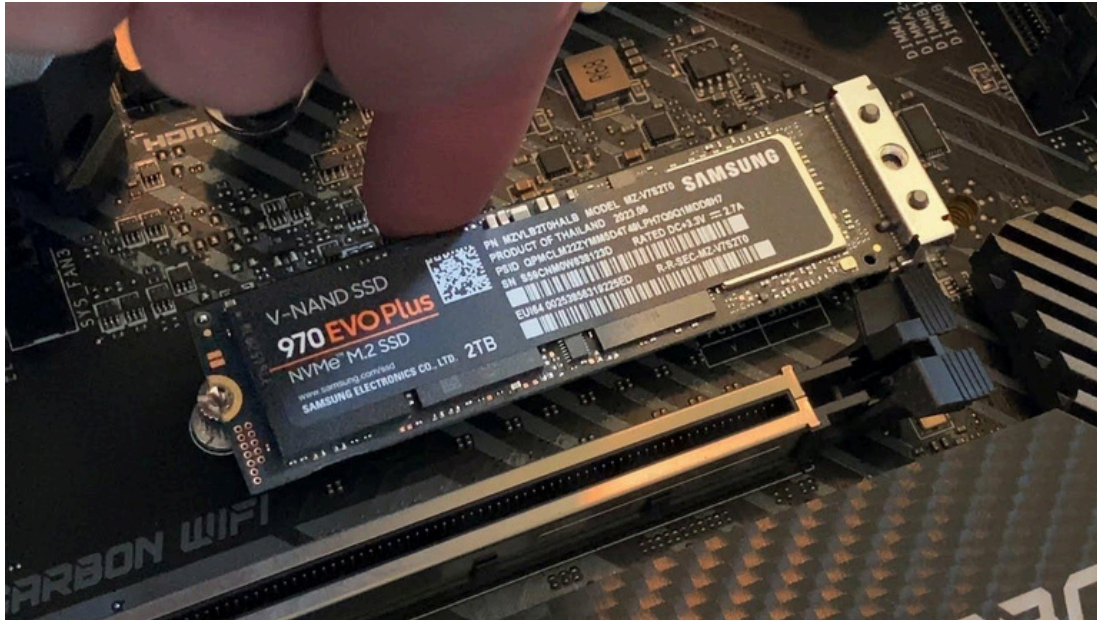
- Shut down the computer and unplug power.
- Use ESD protection or touch an unpainted metal surface before handling components.
- Open the RAM slot clips and remove the module by its edges.
- Align the RAM notch with the slot.
- Press the module down evenly until the clips lock.
- Confirm that the RAM is seated correctly.



Real motherboard installation visual from the supplied assessment.

M.2/NVMe SSD — easy steps

- Switch off and unplug the PC.
- Locate the correct M.2 slot.
- Remove the retaining screw.
- Insert the SSD at a slight angle.
- Press it down gently and secure it with the screw.
- Start the PC and check BIOS/UEFI or Windows to confirm the SSD is detected.



Real M.2 SSD installation photo from the supplied assessment.

Safety: disconnect power, use ESD precautions, hold modules by the edges, and avoid touching gold contacts.

Task 4 — Full Desktop PC Assembly

1. **Prepare the work area** — Switch off and unplug the PSU. Use ESD protection.
2. **Install the CPU** — Align the CPU correctly with the socket marking and lock the socket.
3. **Install the cooler** — Apply thermal paste if required, mount the cooler and connect the CPU fan.
4. **Install RAM** — Align the notch and press until the slot clips lock.
5. **Install storage** — Install M.2/NVMe in the M.2 slot, or connect SATA data + power for a SATA drive.
6. **Install motherboard** — Align it with the case standoffs and rear I/O area, then secure it.
7. **Install GPU** — Insert it into PCIe x16 and secure it; connect required GPU power.
8. **Install PSU** — Secure the PSU and connect 24-pin motherboard + 4/8-pin CPU power.
9. **Connect remaining cables** — Connect storage, GPU, CPU fan and front-panel cables.
10. **Final safety check** — Check seating, loose screws and cable clearance from fans.
11. **Power on** — Connect monitor, keyboard and mains power; start the PC.
12. **Check POST** — Confirm BIOS/UEFI appears and CPU, RAM, storage and GPU are detected.

The build is successful when the system powers on and completes POST without errors.

SECTION C — MINI CAPSTONE PROJECT

Workstation Deployment & Boot-Failure Diagnosis

Objective: Assemble or inspect a workstation, identify likely boot failures, verify compatibility and confirm that the hardware is installed correctly.

1. First three components to check

- PSU — confirm it is switched on and the motherboard/CPU power connectors are connected.
- RAM — confirm the module is seated correctly; loose RAM can cause POST failure or no display.
- CPU + motherboard — confirm the CPU is installed correctly and is supported by the motherboard.

2. Motherboard–CPU compatibility

The CPU must match the motherboard socket and supported CPU generation. Check the motherboard manufacturer's CPU support list and BIOS requirements before installation.

3. Verify RAM and storage

- RAM: confirm the clips are locked and check the correct RAM amount in BIOS/UEFI.
- NVMe: confirm the SSD is seated in the M.2 slot.
- SATA drive: confirm both data and power cables are connected.
- Check BIOS/UEFI first; after Windows starts, verify the drive in Disk Management/File Explorer.

4. Hardware installation checklist

- CPU installed and compatible; cooler connected.
- RAM seated and clips locked.
- SSD/HDD connected and detected.
- 24-pin and 4/8-pin CPU power connected.
- GPU seated and required power connected.
- No loose screws or cables before first boot.
- POST and BIOS/UEFI hardware detection are successful.

5. Why SSD instead of HDD for an office lab?

An SSD gives faster boot times, faster application loading and more responsive multitasking. Because it has no moving mechanical parts, it is also quieter and generally more resistant to physical shock. NVMe SSDs can provide still higher performance through PCIe.

SECTION D — AI-AUGMENTED LEARNING

STEP 1 — BUILD WITH AI

Exact prompt used:

“Act as a junior IT support guide. Create a simple troubleshooting checklist for a newly assembled desktop computer that does not boot. Start with the most basic and important checks, then move to RAM, motherboard, CPU compatibility, storage, and PSU. Include what the technician should check at each step and how BIOS/POST can help identify the problem. Also explain how this checklist can be used to troubleshoot the boot and hardware detection problems reported at ABC Solutions.”

Original AI checklist

- Check the power supply and power connections.
- Check the motherboard for indicator lights or error signals.
- Reseat and test the RAM module.
- Verify CPU and motherboard compatibility.
- Check the storage device and its connections.
- Check BIOS/UEFI to confirm hardware detection.
- Start the system again and verify successful POST.

STEP 2 — TEST & DEBUG (WITHOUT AI)

The checklist was reviewed and reordered so that the safest and most basic physical checks happen first.

Corrected checklist

- Check the PSU and main power connection.
- Check the 24-pin motherboard and CPU power cables.
- Check whether the CPU is correctly installed and compatible.
- Reseat the RAM and check whether it is detected.
- Check the GPU connection if a dedicated GPU is installed.
- Check the storage device and its connections.
- Start the PC and check POST/BIOS for error messages.

What changed and why: Power and motherboard connections were moved to the beginning because the PC cannot start without correct power. CPU, RAM and GPU checks come before storage because these components can prevent POST. BIOS/POST is then used to confirm whether the hardware problem has been resolved.

WEB REFERENCES & REAL-HARDWARE VISUAL SOURCES

These references were checked while preparing the easy explanation and are useful for students who want to see real hardware installation examples.

Crucial — Desktop RAM installation guide

<https://stage-in.crucial.com/content/crucial/en-in/home/support/install-memory-desktop.html>

Crucial — M.2 SSD installation guide

<https://www.crucial.com/content/dam/crucial/ssd-products/ssd-family/documents/installation-guides/crucial-m2-ssd-install-guide-en.pdf>

MSI — Power supply connectors and PSU selection

https://www.msi.com/support/technical_details/MB_SPEC_PSU_Jack

Overclockers UK — Motherboard component layout image

<https://www.overclockers.co.uk/blog/anatomy-of-a-motherboard-everything-you-need-to-know/>

Dell — System board connector examples

<https://www.dell.com/support/manuals/en-in/dell-ebt2250-desktop/dell-tower-plus-ebt2250-owners-manual/installing-the-system-board>

Note: The technical answers in this assignment follow the wording and practical requirements of the uploaded assessment. Web sources are used only to support the visual/installation explanations.